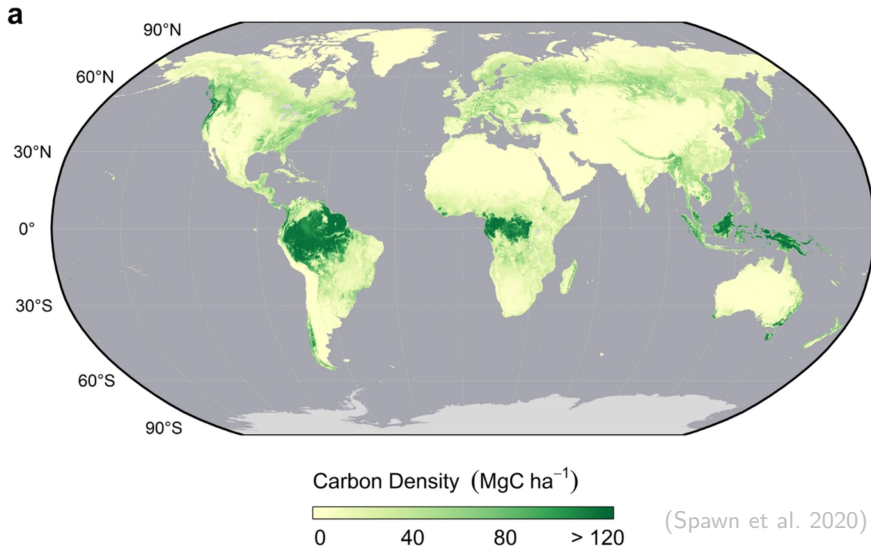


Conservation Policy with Political Resistance

Robin Burgess, LSE
Francisco Costa, FGV EPGE
Allan Hsiao, Stanford
Benjamin Olken, MIT
Veronica Salazar Restrepo, IIES

September 4, 2026

Forest conservation protects major carbon stocks



But conservation is political

N New Internationalist

View from Brazil: Agribusiness lobby scuppers climate gains

Lula wants Brazil to be a beacon in the fight against global warming, Leonardo Sakamoto. The powerful lobby that represents agribusiness in...

Sep 4, 2023



M Mongabay

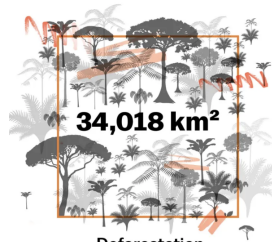
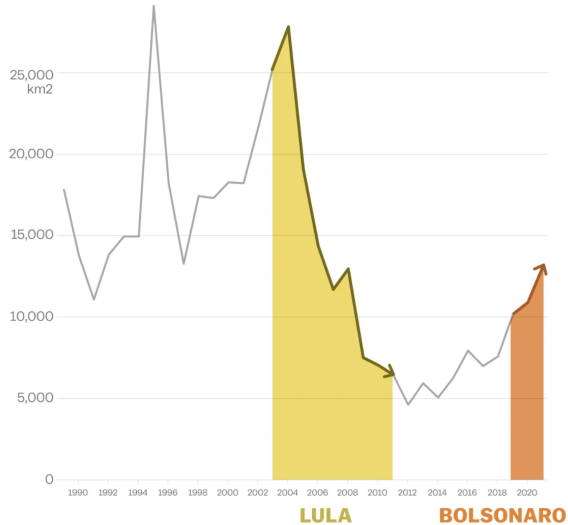
Indonesia palm oil lobby pushes 1 million hectares of new Sulawesi plantations

A state-owned palm oil company and an industry association have begun early work to push a vast new plantation strategy in Sulawesi,...

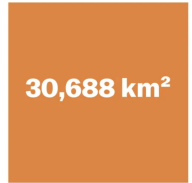
Aug 8, 2024



And politics threatens conservation



Deforestation
under Bolsonaro



Total area
of Belgium

(Northrop 2022)

Question

How can environmental regulation navigate political resistance?

This paper

- ① Regulation faces resistance
 - In Brazil, conservation mobilized agricultural interests
- ② Resistance shapes optimal policy
 - Regulatory burden raises repeal risk
- ③ Optimal policy is targeted
 - Where abatement is high and burden is low

Regulation faces resistance

The Brazilian Amazon

- **PPCDAm** strengthened forest protections beginning in 2004
 - Forest Code requires 80% of each property to remain forested
 - Satellite monitoring, protected areas, and tougher penalties
- Deforestation fell sharply through 2012
 - And then rose again as enforcement weakened

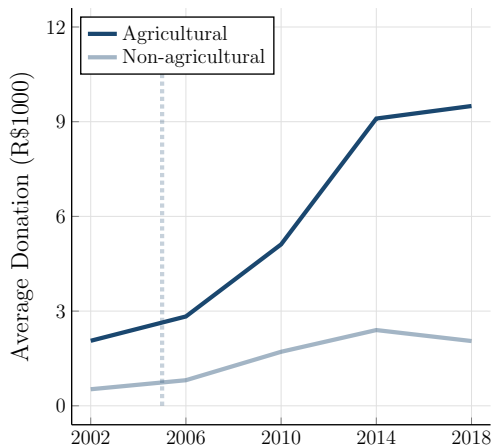
Measuring political resistance

	Donors	Candidates
Full sample	7,337	980
Agricultural firm owners	810	134
Environmental infractors	322	166

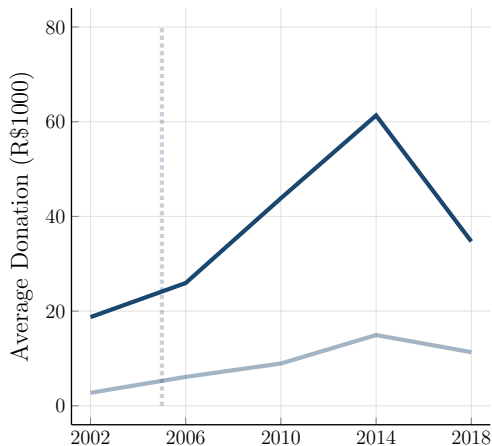
- State and federal legislative elections from 2002 to 2018
 - Universe of formal campaign donations (TSE)
 - Linked to firm ownership (*Receita Federal*)
 - And to environmental infractions (IBAMA)

Agricultural donations over time

Donors



Candidates



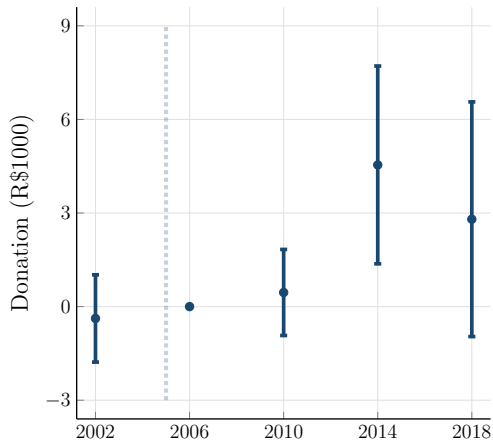
Difference-in-differences

$$y_{imt} = \sum_s \beta_s \text{Agricultural}_i \times \mathbb{1}\{t = s\} + \alpha_i + \gamma_{mt} + \varepsilon_{imt}$$

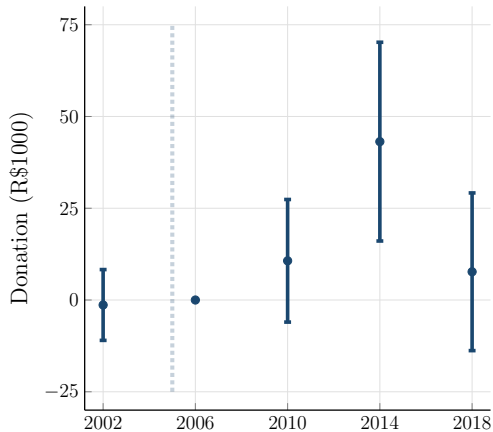
- Agricultural vs. non-agricultural, before vs. after PPCDAm
- Individuals i , municipalities m , election years t
 - Individual and municipality-by-year fixed effects
 - Matching on firm age, structure, size, capital, tax regime (entropy balancing)

PPCDAm mobilized agricultural donations

Donors



Candidates



Agricultural donors gave more

	All candidates	Agricultural candidates
$Ag \times Post\ 2006$	2.79*** (1.02)	2.33*** (0.62)
Effect as % of mean	116	212
Observations	35,195	35,195
Fixed effects	$i + mt$	$i + mt$
Entropy balancing	x	x

Especially where enforcement was binding

Standardized H :	Deforestation reduction	Cloud coverage
$Ag \times Post\ 2006$	1.50** (0.76)	7.41*** (2.58)
$Ag \times Post\ 2006 \times H$	1.82** (0.89)	-11.78** (4.72)
H effect as % of mean	73	-453
Observations	33,615	27,385
Fixed effects	$i + mt$	$i + mt$
Entropy balancing	x	x

Agricultural candidates received more donations

	All donors	Agricultural donors
$Ag \times Post\ 2006$	21.19*** (7.07)	6.58*** (2.12)
Effect as % of mean	95	132
Observations	3,480	3,480
Fixed effects	$i + mt$	$i + mt$
Entropy balancing	x	x

And more votes

	All elections	Federal elections	State elections
$Ag \times Post\ 2006$	10.05*** (2.82)	21.49** (8.47)	1.95* (1.12)
Effect as % of mean	46	47	16
Observations	1,328	260	821
Fixed effects	$i + mt$	$i + mt$	$i + mt$
Entropy balancing	x	x	x

Where is resistance strongest?

$$S'_m = z(\log(S_m)), \quad S_m = \frac{L_m}{N_m} \log\left(1 + \frac{A_m}{F_m}\right)$$

- Economic stakes: implied surplus per producer (donor)
 - Land use reveals agricultural value
 - Pre-period municipal data, logit measure
- Validation
 - Observed census revenue per producer (model-free)
 - Conditioning on carbon density (abatement vs. burden)

Resistance rises with economic stakes

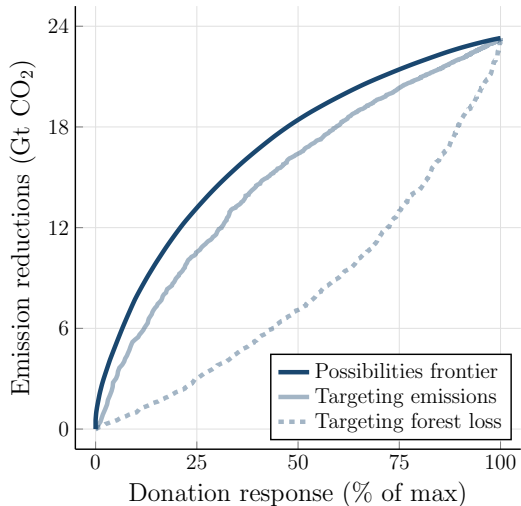
	Implied surplus	Observed revenue
$Ag \times Post\ 2006$	1.83** (0.91)	1.83 (1.12)
$Ag \times Post\ 2006 \times S$	2.75*** (0.93)	2.17* (1.11)
S effect as % of mean	110	87
Observations	33,880	33,880
Fixed effects	$i + mt$	$i + mt$
Entropy balancing	x	x

But not with carbon density

	Implied surplus	Carbon density
$Ag \times Post\ 2006$	1.83** (0.91)	1.79* (0.99)
$Ag \times Post\ 2006 \times S$	2.75*** (0.93)	2.84*** (0.97)
$Ag \times Post\ 2006 \times E$		0.21 (1.26)
S effect as % of mean	110	114
Observations	33,880	33,880
Fixed effects	$i + mt$	$i + mt$
Entropy balancing	x	x

Targeting expands the PPF

- Political possibilities frontier
 - Emissions vs. resistance
 - Each step adds a municipal ban
- Targeting emissions
 - Ignores political resistance
 - Bans where emissions are highest
- Targeting forest loss
 - As priority lists do in practice
 - Bans where resistance is highest



Resistance shapes optimal policy

The Pigouvian benchmark

$$a_i(\tau_i) = \mathbb{1}[v_i > \tau_i], \quad w_i(\tau_i) = (v_i - \kappa e_i)a_i(\tau_i)$$

- Each plot i produces or conserves
 - Production generates agricultural value $v_i > 0$ (fixed)
 - Clearing releases emissions e_i at social cost κ (data)
- Pigouvian tax κe_i maximizes welfare
 - But also imposes substantial producer burden
 - Tax revenue is a transfer from producers to government

Abatement and burden can diverge

$$b_i(\tau_i) = PS_i(0) - PS_i(\tau_i) = \min\{v_i, \tau_i\}$$

	Value	Production	Abatement	Burden
Marginal	$v_i < \tau_i$	Stops	e_i	v_i
Inframarginal	$v_i > \tau_i$	Continues	0	τ_i

Resistance can undo regulation

$$F(\tau) = \underbrace{[1 - R(B(\tau))] W(\tau)}_{\text{regulation}} + \underbrace{R(B(\tau)) W(0)}_{\text{repeal}}$$

- Regulation survives or is repealed
 - Regulation imposes total burden $B(\tau)$
 - Burden raises repeal risk with $R'(B) > 0$
- Optimal policy maximizes **feasible welfare** $F(\tau)$
 - Balances welfare gains against repeal risk

Resistance narrows the Pigouvian tax

$$\tau_i^* = \underbrace{\kappa e_i}_{\text{Pigouvian}} \cdot \mathbb{1} \left[\frac{v_i}{\kappa e_i} < \frac{1}{1 + \mu^*} \right]$$

- Pigouvian with a political cutoff
 - Regulate only where emissions are high relative to value
 - Exempt all efficient and some mildly inefficient production
- Resistance tightens the cutoff
 - Added burden has shadow cost $\mu^* > 0$

Optimal policy is targeted

Quantifying policy impacts

- We build a simple empirical model
 - Characterizes value (targeting)
 - Predicts land use responses (abatement)
 - Predicts producer surplus (burden)
- We then compare optimal and practical policies

Agricultural land use reveals value

$$\max \left\{ v_i - \tau_i + \epsilon_{ij}^a, \epsilon_{ij}^f \right\}, \quad \log \left(\frac{a_i}{f_i} \right) = \frac{v_i - \tau_i}{\sigma}$$

- Site i allocates each hectare j to agriculture or forest
 - Adding logit shocks with scale σ (continuous site-level shares)
- Land use shares reveal value relative to σ

Agricultural returns identify scale

$$\log\left(\frac{a_{it}}{f_{it}}\right) = \beta \underbrace{y_i p_t}_{\text{returns}} + \alpha_i + \gamma_t + \varepsilon_{it}, \quad \sigma = \frac{1}{\beta}$$

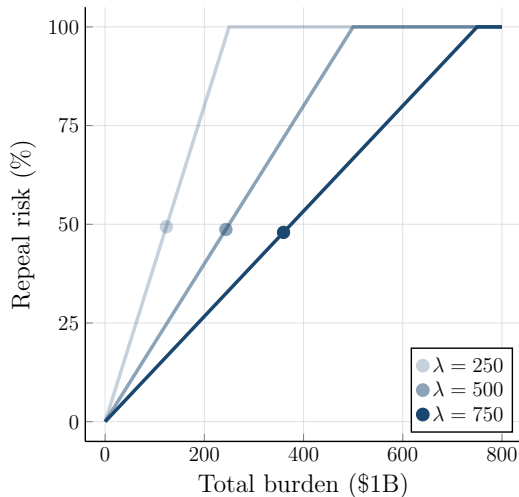
- Coefficient β sets the monetary scale
 - Governs land use responses to returns (and taxes)
 - Gives producer surplus in dollars
- Identification: local crop yields $\times$ global price changes
 - Pooled 10-year long differences capture gradual adjustment
 - Spatial panel of 0.25-degree cells from 2001 to 2019

Agricultural land use responds to returns

	Crop weights: soy, maize, pasture		
	Baseline shares	Leave-one-out shares	Fitted shares
Agricultural returns	0.126*** (0.045)	0.134*** (0.048)	0.128*** (0.049)
Observations	46,682	46,682	46,682
Fixed effects	$iw + tw$	$iw + tw$	$iw + tw$

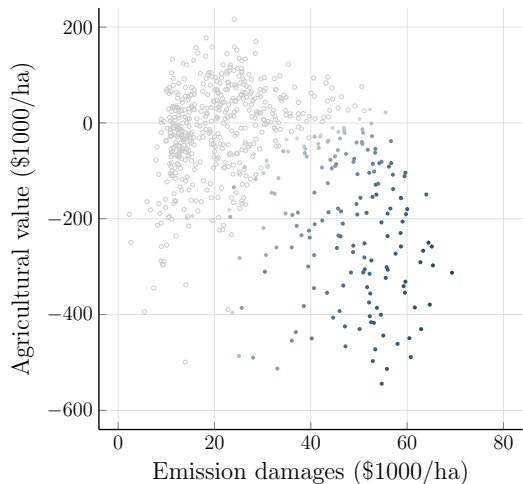
Optimal policy depends on repeal risk

- The land use model gives $B(\tau)$
 - Economic stakes raise donations
- We specify repeal risk $R(B)$
 - Repeal is certain if $B(\tau) > \lambda$
- Dots maximize feasible welfare
 - Higher λ allows greater burden



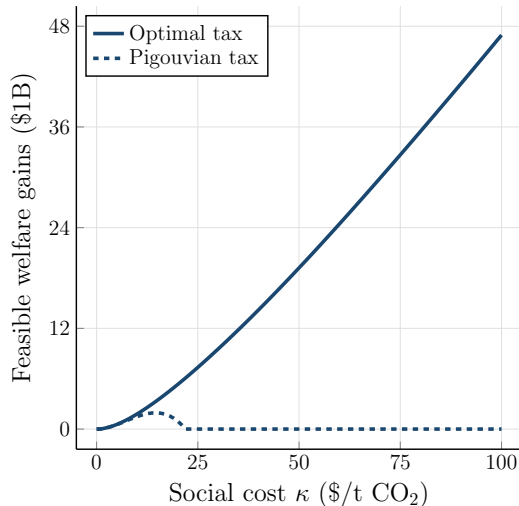
Optimal policy targets both emissions and value

- Pigouvian with political shading
 - Pigouvian helps welfare gains
 - Shading helps repeal risk
- Avoids low- e , high- v munis (lighter)
 - Low welfare gains
 - High repeal risk
- Targets high- e , low- v munis (darker)



Optimal policy avoids repeal

- Pigouvian tax collapses
 - When we account for repeal risk
 - High burden $\rightarrow$ inevitable repeal
- Optimal tax survives
 - With targeted shading
 - Welfare gains grow in κ

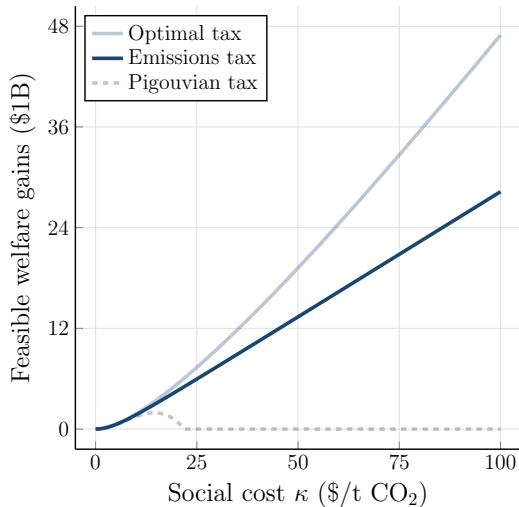


Can practical policies realize the gains from targeting?

- Optimal policy targets emissions and value
- Practical policies use less information
 - Emissions are observable via satellites
 - But value is private information

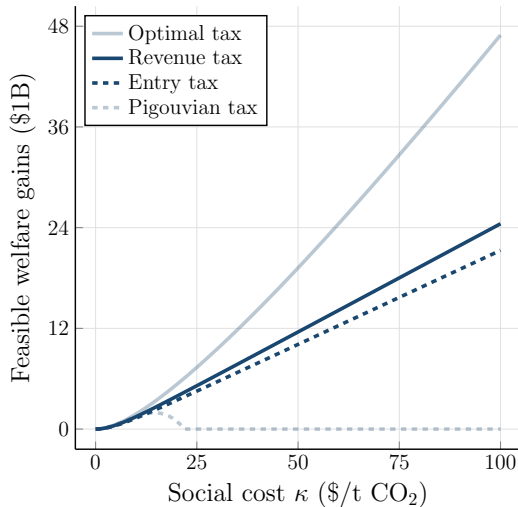
Practical policies capture substantial gains

- Emissions taxes are best
 - Targets e with $\text{corr}(e, v) = -0.36$
 - Reduced tax rate for large κ
- Revenue/entry taxes are good
 - But do not directly target e
- Emissions bans fail, even on the PPF
 - No revenue + producer losses $> \kappa e$



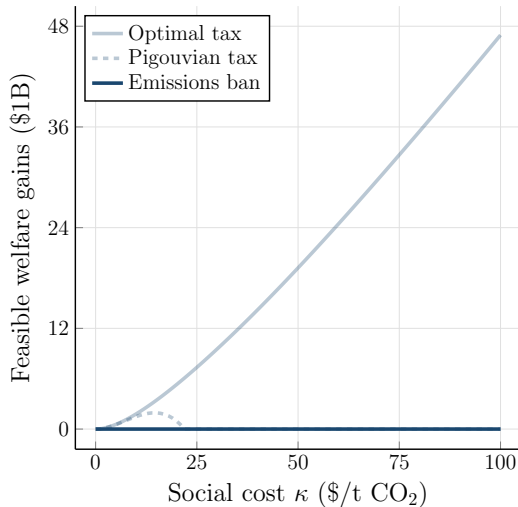
Practical policies capture substantial gains

- Emissions taxes are best
 - Targets e with $\text{corr}(e, v) = -0.36$
 - Reduced tax rate for large κ
- Revenue/entry taxes are good
 - But do not directly target e
- Emissions bans fail, even on the PPF
 - No revenue + producer losses $> \kappa e$



Practical policies capture substantial gains

- Emissions taxes are best
 - Targets e with $\text{corr}(e, v) = -0.36$
 - Reduced tax rate for large κ
- Revenue/entry taxes are good
 - But do not directly target e
- Emissions bans fail, even on the PPF
 - No revenue + producer losses $> \kappa e$



Conclusion

Environmental regulation must be politically feasible

- In Brazil, conservation sparked resistance
 - Agricultural interests responded with money and votes
- Optimal policy is politically targeted